

Frequency Ratio Model for Landslide Susceptibility Mapping of Upper Koyana Region

Abstract: Landslides are one of the most devastating natural hazards, particularly in mountainous terrains such as the Upper Koyana region. This study aims to assess landslide susceptibility using the Frequency Ratio (FR) model by integrating multiple conditioning factors. Twelve key parameters—Relief, Slope, Aspect, Land Use Land Cover (LULC), Profile Curvature, Plan Curvature, Topographic Wetness Index (TWI), Stream Power Index (SRI), Lineament Density, Normalized Difference Vegetation Index (NDVI), Geomorphology, and Rainfall—along with landslide inventory data derived from Google Earth, are analyzed using ArcGIS 10.8. The final output is a Landslide Susceptibility Map that categorizes areas based on their likelihood of experiencing landslides.

1. Introduction:

Landslides pose a significant threat to life, infrastructure, and the environment, especially in the Western Ghats. The Upper Koyana region is particularly prone to landslides due to its rugged topography, heavy rainfall, and geological characteristics. The Frequency Ratio model, a widely used statistical approach, quantifies the relationship between past landslide occurrences and conditioning factors to produce a susceptibility map.

2. Study Area:

The Upper Koyana region, located in the Western Ghats of Maharashtra, India, is characterized by steep slopes, high rainfall, and complex geological formations. The region is crucial due to the presence of the Koyana Dam and surrounding settlements, making landslide risk assessment essential.

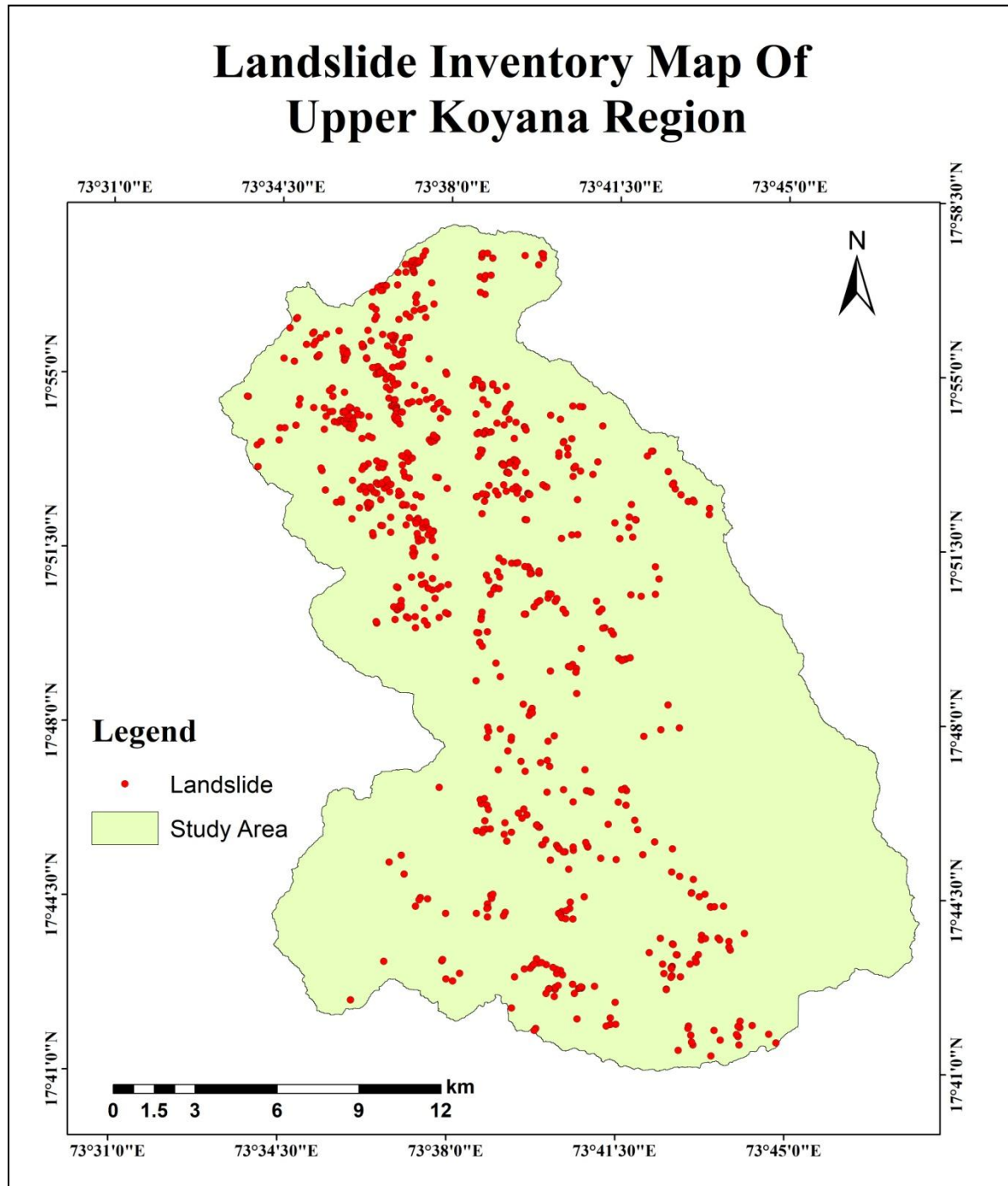
3. Data and Methodology

3.1 Data Sources

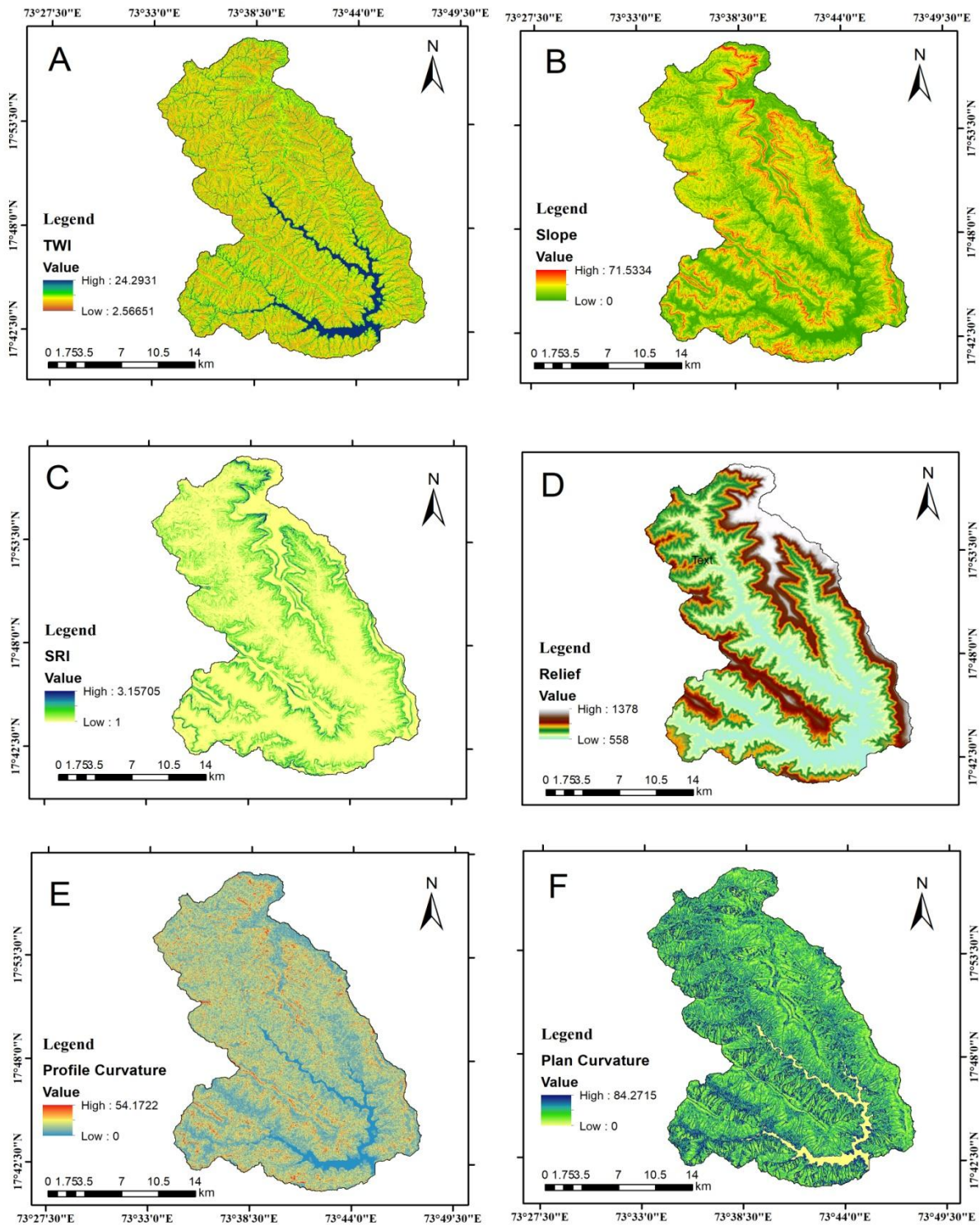
- Landslide Inventory: Mapped using Google Earth imagery and historical records.
- DEM (Digital Elevation Model): Used to derive relief, slope, aspect, and curvature parameters.
- Land Use Land Cover (LULC): Extracted from satellite imagery.
- Rainfall Data: Obtained from IMD.
- Lineament Data: Derived from Bhukosh

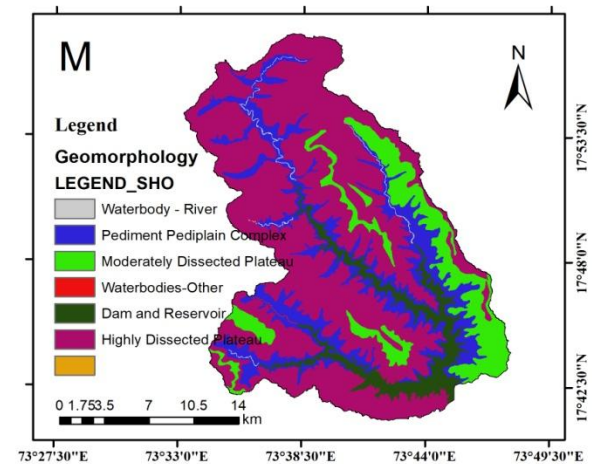
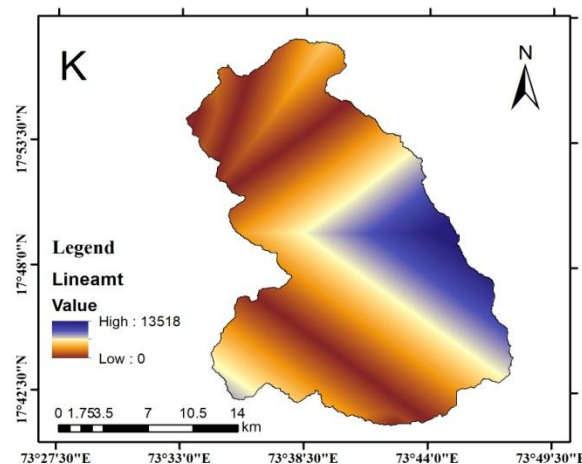
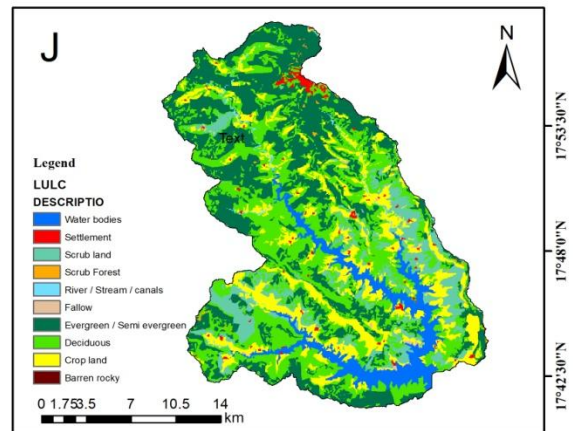
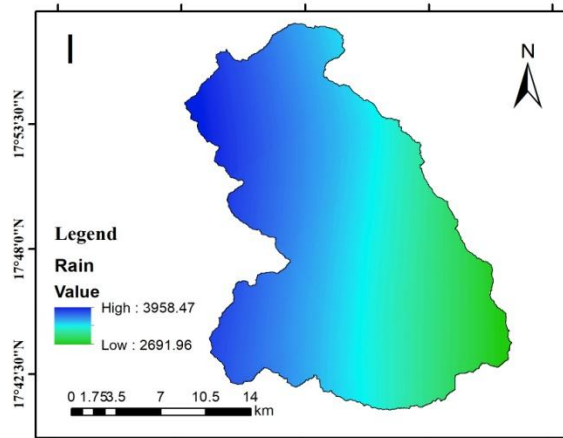
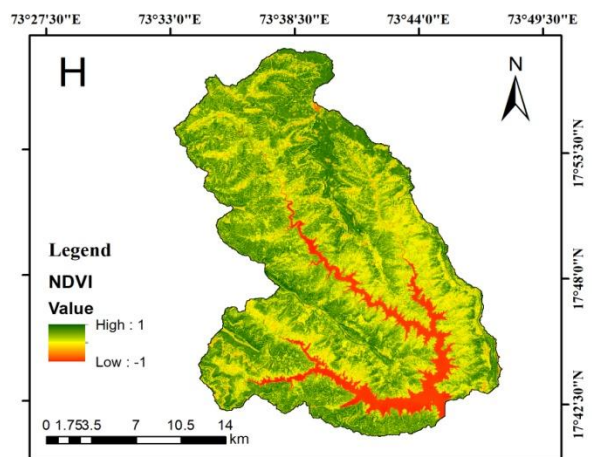
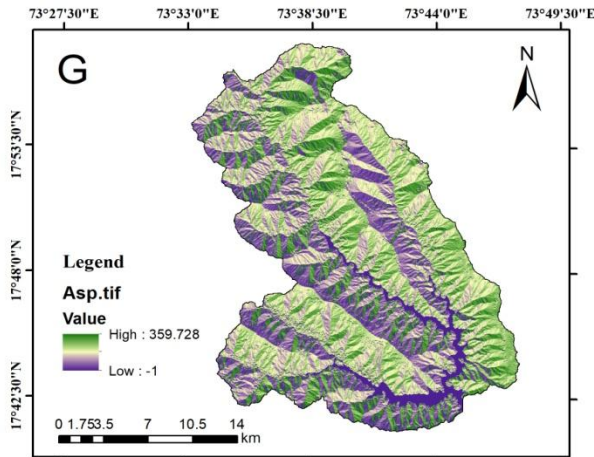
- NDVI: Extracted from satellite images to assess vegetation cover.
- Geomorphology: Classified based on terrain characteristics.

Landslide Inventory Map: Highlights historical landslide occurrences.



Thematic Maps: Individual parameter maps (e.g., slope, aspect, rainfall) illustrate their spatial distribution





A) Topographic Wetness Index (TWI) B) Slope C) Surface Roughness Index (SRI)
D) Relief E) Profile Curvature F) Plan Curvature G) Aspect H) Normalized Difference Vegetation Index (NDVI) I) Rainfall J) Land Use Land Cover (LULC) K) Lineament
M) Geomorphology

3.2 Methodology

1. Preprocessing

- Data collection and conversion to raster format in ArcGIS 10.8.
- Standardization and reclassification of parameter layers.

2. Frequency Ratio Model Calculation

- The FR value is computed as:

$$FR = (\text{Number of landslides in a factor class}) / (\text{Total area of that factor class}) / (\text{Total number of landslides}) / (\text{Total area of the study area})$$

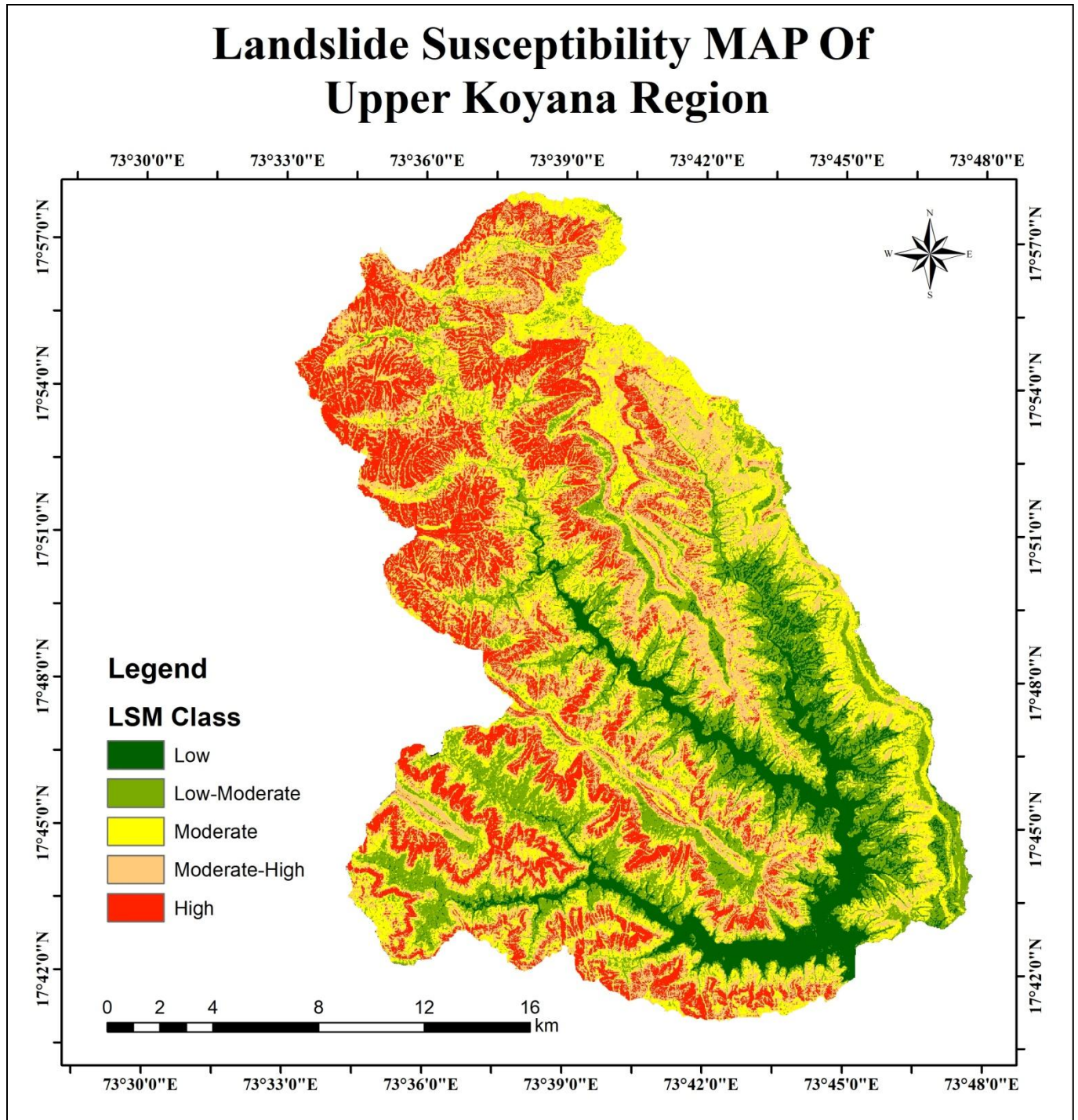
- FR values are calculated for each class of all parameters.

3. Weight Assignment and Susceptibility Mapping

- Each factor is weighted based on its FR value.
- A weighted overlay analysis is performed to generate the final Landslide Susceptibility Map.

4. Results and Discussion

Landslide Susceptibility Map: Classified into low, low - moderate, moderate, moderate- high , and high susceptibility zones.



4. Conclusion:

The study successfully applied the Frequency Ratio model to assess landslide susceptibility in the Upper Koyana region. The integration of 12 key factors allowed for a comprehensive evaluation of landslide-prone areas, highlighting regions at high risk. The final Landslide Susceptibility Map provides crucial insights that can aid in land-use planning, infrastructure development, and disaster preparedness.